

1) In the following question the letters A and B stand for two English sentences. Both A and B are true; but beyond this you are not told anything about them. Can you, from this information alone, determine whether the following longer sentences containing A and B are true or false? Justify your answer in each case, noting any difficulties which may arise. Note that each part of the question is a separate task; there is no assumption that A and B need be the same pair of sentences in each task from (a)-(g), just that in each separate case A and B are true sentences.

- a) 'A' contains more words than 'B'.
- b) A; but B.
- c) Suppose A; then B.
- d) A or B.
- e) The Prime Minister knows that A and B.
- f) A is a necessary and sufficient condition for B.
- g) A only if B.

Solutions:

- 2) In what follows, P and Q stand for English declarative sentences (sentences which have a truth-value, true or false). An argument is a collection of declarative sentences, one of which is marked off from the others (for example, by the word 'therefore') and claimed to follow from the others. The following is a valid form of argument; 'Q is true' follows from the other sentences, whatever declarative sentences we choose for P and Q: P is true; If P, then Q; therefore, Q is true.

Comment on the validity of the following arguments. Where appropriate, give examples to support your reasoning:

- a) Q is true; If P then Q, therefore, P is true.
- b) Peter believes P; If P then Q, therefore, Peter believes Q.
- c) Jane does not believe P; If P then Q; therefore, Jane does not believe Q.
- d) Siobhan believes P. Therefore, if Siobhan believes 'If P then Q', Siobhan believes Q.
- e) Derek does not believe Q. If P then Q. Therefore, Derek ought not to believe P.
- f) P is true. Prudence does not believe 'If P then Q'. Therefore, Prudence does not believe Q.

Solutions:

3) “They all seem to have a motive, sir”, said Lewis, surveying the mangled remains of Jobsworth the Porter, “so the problem is whether the Bursar is telling the truth. I’m assuming of course that everyone is either lying, or telling the truth, but not both.” “Well done, Lewis!”, cried Morse. “In logic there are no surprises. And if the Bursar is telling the truth, so is the Chaplain.” “What’s more, sir”, said Lewis, “the Chaplain and the Dean have given conflicting statements, so they can’t both be telling the truth.” “Excellent, Lewis!”, exclaimed the great detective. “We’ll make a copper of you yet. But nor are the Dean and the Warden both lying. And if the Warden is telling the truth, the Chaplain isn’t.” “I think I follow that, sir. But we still don’t know if the bursar is lying.”

Giving the clearest reasoning for your conclusion possible, assess whether Lewis’s final claim, ‘But we still don’t know if the bursar is lying’, is true, if we accept the truth of the other claims made by Lewis and Morse.

Solution: